```
import shutil
import sys

import torch

print("Python :", sys.version.split()[0])
print("Torch  :", torch.__version__, "| CUDA build:", torch.version.cuda)

assert torch.cuda.is_available(), (
    "GPU belum aktif. Runtime > Change runtime type > Hardware accelerator > A100 GPU, "
    "lalu jalankan ulang dari Langkah 1."
)

name = torch.cuda.get_device_name(0)
prop = torch.cuda.get_device_properties(0)
vram_gb = prop.total_memory / 1024**3
cc = torch.cuda.get_device_capability(0)
bf16 = torch.cuda.is_bf16_supported()

print()
print("GPU    :", name)
print(f"VRAM   : {vram_gb:.1f} GB")
print(f"CC     : {cc[0]}.{cc[1]}  -> bf16 native: {bf16}")

assert cc[0] >= 8 and vram_gb >= 35, (
    f"Notebook ini disetel untuk A100, runtime sekarang: {name} "
    f"({vram_gb:.1f} GB, CC {cc[0]}.{cc[1]}).\n"
    "Pindah ke A100 lewat Runtime > Change runtime type > A100 GPU, lalu Run all.\n"
    "Kalau harus tetap di T4: ubah Langkah 5 menjadi amp_dtype float16 dan IMG_SIZE 840."
)

free_gb = shutil.disk_usage("/content").free / 1024**3
print(f"Disk   : {free_gb:.1f} GB bebas di /content (butuh ±4.5 GB)")
assert free_gb > 6, "disk /content tidak cukup, hapus file lama atau ganti runtime"
```

```
Python : 3.13.16
Torch  : 2.11.0+cu130 | CUDA build: 13.0

GPU    : NVIDIA A100-SXM4-40GB
VRAM   : 39.5 GB
CC     : 8.0  -> bf16 native: True
Disk   : 234.6 GB bebas di /content (butuh ±4.5 GB)
```

```
!test -d /content/count-anything/.git || git clone --depth 1 https://github.com/Mengqi-Lei/count-anything.git /content/
%cd /content/count-anything
!git log -1 --pretty='commit %h %s'
```

```
/content/count-anything
commit 9ab1d5a Update README_CN.md
```

```
%%bash
set -eo pipefail
cd /content/count-anything

grep -v -E '^(torch|torchvision|triton|decord|fvcore|fairscale)([<>=!~].*)?$' requirements.txt > /tmp/req_colab.txt
echo "paket yang diinstal:"; paste -sd' ' /tmp/req_colab.txt; echo

# Repo menulis `opencv-python` dan `pandas>=2.0.0` tanpa batas atas. Rilis terbaru keduanya
# (opencv-python 5.x, pandas 3.x) menuntut numpy>=2, sehingga resolver pip memutar mundur
# mencari versi lama — lama, dan kalau gagal, numpy 2.x tetap terpakai. Pin langsung:
#   opencv-python 4.11 -> numpy>=1.26 (cocok dengan python 3.12)
#   pandas 2.3        -> numpy>=1.26
pip install --no-input -r /tmp/req_colab.txt \
  "numpy==1.26.4" "opencv-python==4.11.0.86" "pandas<3" 2>&1 | tail -n 30

echo
echo "numpy yang benar-benar ada di disk (interpreter baru):"
python -c "import numpy; print(numpy.__version__, numpy.__file__)"
```

```
paket yang diinstal:
timm>=1.0.17 einops>=0.7.0 numpy>=1.26,<2 Pillow>=9.0.0 matplotlib>=3.7.0 pandas>=2.0.0 PyYAML>=6.0 tqdm>=4.60.0 hydra-c

Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.13/dist-packages (from huggingface_hub->-r /tm
Requirement already satisfied: anyio in /usr/local/lib/python3.13/dist-packages (from httpx<1,>=0.23.0->huggingface_hub-
Requirement already satisfied: certifi in /usr/local/lib/python3.13/dist-packages (from httpx<1,>=0.23.0->huggingface_hu
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.13/dist-packages (from httpx<1,>=0.23.0->huggingf
Requirement already satisfied: idna in /usr/local/lib/python3.13/dist-packages (from httpx<1,>=0.23.0->huggingface_hub->
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.13/dist-packages (from httpcore==1.*->httpx<1,>=0.23.
```

```
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.13/dist-packages (from python-dateutil>=2.8.2->pandas<
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.13/dist-packages (from torch->timm>=1.0.17->-r /t
Requirement already satisfied: jinja2 in /usr/local/lib/python3.13/dist-packages (from torch->timm>=1.0.17->-r /tmp/req_
Requirement already satisfied: cuda-toolkit==13.0.2 in /usr/local/lib/python3.13/dist-packages (from cuda-toolkit[cublas
Requirement already satisfied: cuda-bindings<14,>=13.0.3 in /usr/local/lib/python3.13/dist-packages (from torch->timm>=1
Requirement already satisfied: nvidia-cudnn-cu13==9.19.0.56 in /usr/local/lib/python3.13/dist-packages (from torch->timm
Requirement already satisfied: nvidia-cusparselt-cu13==0.8.0 in /usr/local/lib/python3.13/dist-packages (from torch->tim
Requirement already satisfied: nvidia-nccl-cu13==2.28.9 in /usr/local/lib/python3.13/dist-packages (from torch->timm>=1.
Requirement already satisfied: nvidia-nvshmem-cu13==3.4.5 in /usr/local/lib/python3.13/dist-packages (from torch->timm>=
Requirement already satisfied: triton==3.6.0 in /usr/local/lib/python3.13/dist-packages (from torch->timm>=1.0.17->-r /t
Requirement already satisfied: nvidia-cublas==13.1.0.3.* in /usr/local/lib/python3.13/dist-packages (from cuda-toolkit[c
Requirement already satisfied: nvidia-cuda-runtime==13.0.96.* in /usr/local/lib/python3.13/dist-packages (from cuda-tool
Requirement already satisfied: nvidia-cufft==12.0.0.61.* in /usr/local/lib/python3.13/dist-packages (from cuda-toolkit[c
Requirement already satisfied: nvidia-cufile==1.15.1.6.* in /usr/local/lib/python3.13/dist-packages (from cuda-toolkit[c
Requirement already satisfied: nvidia-cuda-cupti==13.0.85.* in /usr/local/lib/python3.13/dist-packages (from cuda-toolki
Requirement already satisfied: nvidia-curand==10.4.0.35.* in /usr/local/lib/python3.13/dist-packages (from cuda-toolkit[
Requirement already satisfied: nvidia-cusolver==12.0.4.66.* in /usr/local/lib/python3.13/dist-packages (from cuda-toolki
Requirement already satisfied: nvidia-cusparse==12.6.3.3.* in /usr/local/lib/python3.13/dist-packages (from cuda-toolkit
Requirement already satisfied: nvidia-nvjitlink==13.0.88.* in /usr/local/lib/python3.13/dist-packages (from cuda-toolkit
Requirement already satisfied: nvidia-cuda-nvrtc==13.0.88.* in /usr/local/lib/python3.13/dist-packages (from cuda-toolki
Requirement already satisfied: nvidia-nvtx==13.0.85.* in /usr/local/lib/python3.13/dist-packages (from cuda-toolkit[cubl
Requirement already satisfied: cuda-pathfinder~=1.1 in /usr/local/lib/python3.13/dist-packages (from cuda-bindings<14,>=
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.13/dist-packages (from sympy>=1.13.3->torch-
Requirement already satisfied: markupsafe>=2.1.1 in /usr/local/lib/python3.13/dist-packages (from werkzeug>=1.0.1->tensc

numpy yang benar-benar ada di disk (interpreter baru):
1.26.4 /usr/local/lib/python3.13/dist-packages/numpy/__init__.py
```

```
!pip install decord
```

```
Requirement already satisfied: decord in /usr/local/lib/python3.13/dist-packages (0.6.0)
Requirement already satisfied: numpy>=1.14.0 in /usr/local/lib/python3.13/dist-packages (from decord) (1.26.4)
```

```
!pip install --upgrade "jax[cpu]>=0.4.26"
```

```
Requirement already satisfied: jax>=0.4.26 in /usr/local/lib/python3.13/dist-packages (from jax[cpu]>=0.4.26) (0.12.0.1)
Requirement already satisfied: jaxlib<=0.12.0.1,>=0.11.2 in /usr/local/lib/python3.13/dist-packages (from jax>=0.4.26->j
Requirement already satisfied: ml_dtypes>=0.5.0 in /usr/local/lib/python3.13/dist-packages (from jax>=0.4.26->jax[cpu]>=
Collecting numpy>=2.2 (from jax>=0.4.26->jax[cpu]>=0.4.26)
  Using cached numpy-2.5.4-cp313-cp313-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (6.6 kB)
Requirement already satisfied: opt_einsum in /usr/local/lib/python3.13/dist-packages (from jax>=0.4.26->jax[cpu]>=0.4.26
Requirement already satisfied: scipy>=1.15 in /usr/local/lib/python3.13/dist-packages (from jax>=0.4.26->jax[cpu]>=0.4.2
Using cached numpy-2.5.4-cp313-cp313-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl (16.7 MB)
Installing collected packages: numpy
  Attempting uninstall: numpy
    Found existing installation: numpy 1.26.4
    Uninstalling numpy-1.26.4:
      Successfully uninstalled numpy-1.26.4
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviou
numba 0.64.0 requires numpy<2.5,>=1.22, but you have numpy 2.5.4 which is incompatible.
Successfully installed numpy-2.5.4
WARNING: The following packages were previously imported in this runtime:
  [numpy]
You must restart the runtime in order to use newly installed versions.
```

RESTART SESSION

```
!pip install --force-reinstall "numpy==1.26.4"
```

```
Collecting numpy==1.26.4
  Using cached numpy-1.26.4-cp313-cp313-linux_x86_64.whl
Installing collected packages: numpy
  Attempting uninstall: numpy
    Found existing installation: numpy 2.5.4
    Uninstalling numpy-2.5.4:
      Successfully uninstalled numpy-2.5.4
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviou
jax 0.12.0.1 requires numpy>=2.2, but you have numpy 1.26.4 which is incompatible.
jaxlib 0.12.0.1 requires numpy>=2.0, but you have numpy 1.26.4 which is incompatible.
opencv-contrib-python 4.14.0.94 requires numpy>=2; python_version >= "3.9", but you have numpy 1.26.4 which is incompati
tobler 0.14.0 requires numpy>=2.0, but you have numpy 1.26.4 which is incompatible.
xarray-einstats 0.11.0 requires numpy>=2.0, but you have numpy 1.26.4 which is incompatible.
shap 0.54.1 requires numpy>=2, but you have numpy 1.26.4 which is incompatible.
geopandas 1.2.0 requires numpy>=2, but you have numpy 1.26.4 which is incompatible.
cupy-cuda13x 14.0.1 requires numpy<2.6,>=2.0, but you have numpy 1.26.4 which is incompatible.
ml-dtypes 0.6.0 requires numpy>=2.0.0, but you have numpy 1.26.4 which is incompatible.
ml-dtypes 0.6.0 requires numpy>=2.1.0; python_version >= "3.13", but you have numpy 1.26.4 which is incompatible.
pytensor 2.38.3 requires numpy>=2.0, but you have numpy 1.26.4 which is incompatible.
dm-tree 0.1.10 requires numpy>=2.1.0; python_version >= "3.13", but you have numpy 1.26.4 which is incompatible.
rasterio 1.5.2 requires numpy>=2, but you have numpy 1.26.4 which is incompatible.
opencv-python-headless 5.0.0.93 requires numpy>=2; python_version >= "3.9", but you have numpy 1.26.4 which is incompati
Successfully installed numpy-1.26.4
WARNING: The following packages were previously imported in this runtime:
  [numpy]
You must restart the runtime in order to use newly installed versions.
```

RESTART SESSION

```
import importlib
import subprocess
import sys

import numpy as np

ondisk = subprocess.run(
    [sys.executable, "-c", "import numpy; print(numpy.__version__)"],
    capture_output=True, text=True,
).stdout.strip()

print("numpy di kernel :", np.__version__, "|", np.__file__)
print("numpy di disk   :", ondisk)

if not ondisk.startswith("1."):
    raise AssertionError(
        "numpy di disk masih 2.x — sel install di atas belum berhasil menurunkannya. "
        "Jalankan ulang Langkah 3 tanpa -q untuk melihat pesan pip, atau paksa: "
        "!pip install --no-deps 'numpy==1.26.4'"
    )

if not np.__version__.startswith("1."):
    raise AssertionError(
        "numpy di disk SUDAH 1.x, tapi kernel notebook masih memegang 2.x dari sebelum install. "
        "Lakukan Runtime > Restart session, lalu jalankan ulang sel ini "
        "(sel install tidak perlu diulang kalau tidak ada paket yang berubah)."
    )

MODULES = [
    "timm", "einops", "yaml", "hydra", "omegaconf", "cv2", "pycocotools",
    "skimage", "sklearn", "scipy", "torchmetrics", "triton", "huggingface_hub", "ftfy",
]
failed = []
for module in MODULES:
    try:
        importlib.import_module(module)
    except Exception as exc:
        failed.append(f"{module}: {type(exc).__name__}: {exc}")
print("gagal import  :", failed or "No")

sys.path.insert(0, "/content/count-anything")
from count_anything import CountAnything

print("import package:", "CountAnything OK")
```

```
numpy di kernel : 1.26.4 | /usr/local/lib/python3.13/dist-packages/numpy/__init__.py
numpy di disk   : 1.26.4
gagal import  : No
import package: CountAnything OK
```

```
from pathlib import Path

from huggingface_hub import hf_hub_download

CKPT = Path("/content/count-anything/checkpoints/count_anything.pt")
CKPT.parent.mkdir(parents=True, exist_ok=True)
MIN_BYTES = 3_500_000_000  # 3.57 GB

if CKPT.exists() and CKPT.stat().st_size >= MIN_BYTES:
    print("checkpoint sudah lengkap, unduh dilewati")
else:
    CKPT.unlink(missing_ok=True)
    hf_hub_download(
        repo_id="MengqiLei/count-anything",
        filename="count_anything.pt",
        local_dir=str(CKPT.parent),
    )

size = CKPT.stat().st_size
print(f"{CKPT} -> {size / 1e9:.2f} GB")
assert size >= MIN_BYTES, "ukuran checkpoint tidak sesuai (unduhan terpotong), ulangi sel ini"
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOK
count_anything.pt: reconstructing file: 100%                    3.57GB / 3.57GB,  224MB/s
count_anything.pt: downloading bytes:                           3.38GB,  222MB/s
/content/count-anything/checkpoints/count_anything.pt -> 3.57 GB
```

```
from pathlib import Path

import torch
import yaml

SRC_CFG = Path("/content/count-anything/config/count_anything_test_cloc.yaml")
DST_CFG = Path("/content/count-anything/config/count_anything_inference_colab.yaml")

IMG_SIZE = 1008
AMP_DTYPE = "bfloat16"

cc = torch.cuda.get_device_capability(0)
assert cc[0] >= 8, (
    f"bf16 native butuh CC >= 8.0, GPU sekarang CC {cc[0]}.{cc[1]}. "
    "Jalankan Langkah 1 / pindah ke A100."
)
assert torch.cuda.is_bf16_supported(), "runtime melaporkan bf16 tidak didukung"

cfg = yaml.safe_load(SRC_CFG.read_text())
amp = cfg["trainer"]["optim"]["amp"]
amp["enabled"] = True
amp["amp_dtype"] = AMP_DTYPE

torch.backends.cudnn.benchmark = True
torch.backends.cuda.matmul.allow_tf32 = True

DST_CFG.write_text(yaml.safe_dump(cfg, sort_keys=False, allow_unicode=True))
print("config   :", DST_CFG.name)
print("amp      :", amp["amp_dtype"], "| enabled:", amp["enabled"])
print("resolusi :", IMG_SIZE, "(= resolusi training, tidak diubah)")
```

```
config   : count_anything_inference_colab.yaml
amp      : bfloat16 | enabled: True
resolusi : 1008 (= resolusi training, tidak diubah)
```

```
USE_UPLOAD = False
SAMPLE_URL = "http://images.cocodataset.org/val2017/000000039769.jpg"

import urllib.request
from pathlib import Path

if USE_UPLOAD:
    from google.colab import files

    uploaded = files.upload()
    IMAGE_PATH = Path("/content") / next(iter(uploaded))
else:
    IMAGE_PATH = Path("/content/sample_cats.jpg")
```

```
    if not IMAGE_PATH.exists():
        urllib.request.urlretrieve(SAMPLE_URL, IMAGE_PATH)

from PIL import Image as PILImage

width, height = PILImage.open(IMAGE_PATH).size
print(f"{IMAGE_PATH}  ({width}x{height}, {IMAGE_PATH.stat().st_size / 1024:.0f} KB)")
```

```
/content/sample_cats.jpg  (640x480, 169 KB)
```

```
import time
import subprocess
import sys
import torch
from pathlib import Path

sys.path.insert(0, "/content/count-anything")

from count_anything import CountAnything
from IPython.display import Image as DisplayImage
from IPython.display import display

IMAGE_PATH = Path("/content/container.webp")
QUERIES = ["container"]

model = CountAnything(
    checkpoint="checkpoints/count_anything.pt",
    config="config/count_anything_inference_colab.yaml",
    output_dir="exp/count_anything_inference",
    num_gpus=1,
)

for query in QUERIES:
    started = time.perf_counter()
    try:
        def verbose_run(config_path):
            import os
            import subprocess
            env = os.environ.copy()
            env["PYTHONPATH"] = str(model.repo_root) + os.pathsep + env.get("PYTHO

            mock_script = f"""import sys, types
tb_mock = types.ModuleType('tensorboard')
tb_writer_mock = types.ModuleType('writer')
class SummaryWriter:
    def __init__(self, *args, **kwargs): pass
    def add_scalar(self, *args, **kwargs): pass
    def close(self, *args, **kwargs): pass
tb_writer_mock.SummaryWriter = SummaryWriter
sys.modules['torch.utils.tensorboard'] = tb_writer_mock
sys.modules['tensorboard'] = tb_mock

# Completely mock fvcore param schedulers
fv_mock = types.ModuleType('fvcore')
fv_common_mock = types.ModuleType('common')
fv_param_mock = types.ModuleType('param_scheduler')
class MockScheduler:
    def __init__(self, *args, **kwargs): pass
    def __call__(self, *args, **kwargs): return 1.0
fv_param_mock.ConstantParamScheduler = MockScheduler
fv_param_mock.MultiStepParamScheduler = MockScheduler
fv_param_mock.CosineParamScheduler = MockScheduler
sys.modules['fvcore'] = fv_mock
sys.modules['fvcore.common'] = fv_common_mock
sys.modules['fvcore.common.param_scheduler'] = fv_param_mock

import runpy
sys.argv = ['count_anything.train.train', '-c', '{config_path}', '--use-cluster',
runpy.run_module('count_anything.train.train', run_name='__main__', alter_sys=True)
"""

            result = subprocess.run(
                [
                    model.python_executable,
                    "-c",
                    mock_script
                ],
                env=env,
                capture_output=True,
```

```
                text=True,
                check=True,
            )
            return result

        model._run = verbose_run

        result = model(str(IMAGE_PATH), query)[0]
        elapsed = time.perf_counter() - started
        print(f"{query!r} -> {result.count} instance  ({elapsed:.1f}s)")
        saved = result.save()
        print("  gambar:", saved)
        print("  json  :", saved.replace(".jpg", ".json"))
        display(DisplayImage(saved, width=800))
    except subprocess.CalledProcessError as e:
        print("\n=== DETAILED SUBPROCESS ERROR LOGS ===")
        print("STDOUT:\n", e.stdout)
        print("STDERR:\n", e.stderr)
        print("======================================\n")
        raise e

print("VRAM puncak sesi ini:", f"{torch.cuda.max_memory_allocated(0) / 1024**3:.2f}
```

```
'container' -> 64 instance  (28.1s)
  gambar: /content/count-anything/exp/count_anything_inference/container__container__20261106_094056/container__containe
  json  : /content/count-anything/exp/count_anything_inference/container__container__20261106_094056/container__containe
```




```
import zipfile
from pathlib import Path

from google.colab import files

RUNS_DIR = Path("/content/count-anything/exp/count_anything_inference")
runs = sorted(p for p in RUNS_DIR.glob("*") if p.is_dir())
print(f"{len(runs)} run tersimpan, 5 terakhir:")
for run in runs[-5:]:
    print("  -", run.name)

ZIP_PATH = Path("/content/count_anything_results.zip")
with zipfile.ZipFile(ZIP_PATH, "w", zipfile.ZIP_DEFLATED) as archive:
    for path in RUNS_DIR.rglob("*"):
        if path.is_file() and path.suffix in {".jpg", ".json"}:
            archive.write(path, path.relative_to(RUNS_DIR))
print(f"zip    : {ZIP_PATH} ({ZIP_PATH.stat().st_size / 1024:.0f} KB)")

files.download(str(ZIP_PATH))
```